

anchovy

anchovy takes single-cell viral sequencing data and works out, for each individual cell, what the virus's genome looked like in that cell — then compares those genomes across cells to find mutations and how they relate to each other.

You give it sequencing reads that have already been mapped to a reference genome. anchovy sorts the reads by which cell they came from, builds a consensus genome for each cell, and produces tables describing the mutations it found and how the different genotypes are related.

You can run the whole thing as one automated pipeline (recommended), or run each step by hand.

What the pipeline does, step by step

1. **extract** — sorts the reads by the cell barcode they carry
2. **fasta** — writes out the reads for each cell into its own file
3. **map** — lines up each cell's reads against the reference genome
4. **cell consensus** — builds one consensus genome per cell from its reads
5. **merge** — collects all the per-cell genomes into one file
6. **consensus** — trims to the region you care about and lists each cell's mutations
7. **annotate** — works out which mutations change the protein, and builds tables showing how genotypes relate

The final results are spreadsheet-style files (CSV) you can open in Excel or load into other analysis tools. Two of them describe the **genotype network** — how the different viral genotypes relate to one another — and can be opened directly in Cytoscape to view and explore the network visually (see below).

Setting it up

anchovy relies on a few external programs (for mapping and handling sequence files) plus its own code. The easiest way to get everything at once is with **conda**, an environment manager widely used in bioinformatics. If you don't have conda, install [Miniconda](#) first.

Setup is two commands. The first creates a self-contained environment with all the required programs; the second installs anchovy itself into it:

```
bash
```

```
# 1. create the environment (this may take a few minutes)
conda env create -f environment.yml
conda activate anchovy

# 2. install anchovy
pip install -e .
```

Check that it worked:

```
bash

anchovy --version
snakemake --version
```

(If you're curious why it's two steps rather than one: the first command installs the ready-made external programs, and the second installs anchovy's own code in a way that lets you edit it. They're handled differently, so they're separate commands. You don't need to understand the details to use it.)

Running the whole pipeline

You control a run through a **settings file** — a plain text file that lists where your data is and a few options. You never need to edit anchovy's code; you just point it at a settings file. An example, `workflow/config.yaml`, is included.

To run everything:

```
bash

snakemake -s workflow/Snakefile --configfile workflow/config.yaml --cores 8
```

`--cores 8` lets it use 8 processor cores to work on multiple cells at once — set this to however many your machine has.

Tip: before a real run, add `-n` to do a *preview*. This shows you exactly what it would do without actually doing it, so you can check everything looks right first:

```
bash

snakemake -s workflow/Snakefile --configfile workflow/config.yaml --cores 8 -n
```

Another handy feature: if a run stops partway (or you change one setting), you can just run it again and it will pick up where it left off, redoing only the steps that actually need it — not starting over from scratch.

Pointing it at your own data

Copy the example settings file and change the paths to match your data:

```
yaml

sample: "my_sample"           # your reads should be at {data_dir}/{sample}.sam
data_dir: "path/to/data"
whitelist: "path/to/10x_whitelist.txt"  # your list of valid cell barcodes
signature: "CTACACGACGCTCTTCGGATCT..." # the barcode signature for your 10X kit
template: "path/to/reference.fasta"     # the reference genome
reference_name: "your_ref_name"         # the name written after ">" in that file
orf_start: 96                          # start of the region you want analysed
orf_end: 10272                          # end of that region
```

Then run with your file instead of the example:

```
bash

snakemake -s workflow/Snakefile --configfile my_settings.yaml --cores 8
```

If you already have per-cell sequence files

If you've already split your reads into one file per cell (from an earlier run or another tool), you can skip the first steps. Add this line to your settings file pointing at that folder:

```
yaml

cells_dir: "path/to/your/per-cell-files"
```

The included `workflow/config_test.yaml` is a working example of this, set up to run against the small test dataset that ships with anchovy.

Running one step at a time

If you'd rather run steps individually instead of the full pipeline, each is its own command:

```
bash

anchovy extract  reads.sam whitelist.txt -o out_anchovy.csv
anchovy fasta    out_anchovy.csv cells/
anchovy consensus allConsensus.fasta 96 10272 --out-prefix results/sample
anchovy annotate  filtConsensus.csv reference.txt results/sample
```

Add `--help` to any command (e.g. `anchovy extract --help`) to see its options.

Viewing the genotype networks

Two of anchovy's output files describe how the viral genotypes relate to each other as a network:

- `<sample>_genotypeNetwork.csv` — all the relationships between genotypes
- `<sample>_epistaticNetwork.csv` — just the "single-step" links (genotypes that differ by exactly one mutation), plus links back to the reference

You can explore these visually in **Cytoscape**, a free tool for viewing and analyzing networks that's widely used in biology. Download it from cytoscape.org, or use the no-install browser version at web.cytoscape.org.

To load an anchovy network:

1. In Cytoscape, choose **File → Import → Network from File** and pick one of the network CSVs.
2. Cytoscape will recognize the `source` and `target` columns automatically and draw the network — no manual setup needed. The other columns (`overlap`, `mutNumSource`, `count`, and so on) come in as **edge attributes**, properties of each link you can use for styling: for example, make links thicker when more mutations are shared, or color nodes by how many mutations a genotype carries.

Each point (node) in the resulting picture is a genotype; each line (edge) is a relationship between two genotypes. This is a quick way to see, for instance, which mutations tend to build on one another.

For developers

anchovy comes with an automated test suite that checks each step still produces the expected results. To run it:

```
bash
pytest                                # run all tests
pytest tests/test_annotate.py -v      # run the tests for one step
```

Test data lives in `tests/data/`. The `tests/make_*_fixtures.py` scripts regenerate it if ever needed.

Good to know

- The mapping/consensus step uses **sam2consensus**, a small existing tool by Edgardo Ortiz ([original here](#)). A copy, updated to run on modern Python, is included in `workflow/scripts/`.

- The original version of this analysis also made plots. This version produces the data tables only; you can make figures from those in your tool of choice.
- anchovy currently assumes you're mapping against a single reference sequence. A segmented genome (several reference pieces) would need a small extension.

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